

RC circuit laboratory protocol

Synthetic teaching dataset authored for ResearchHub; no real participants.

Objective: estimate the discharge time constant and compare with R times C .

Circuit: a 3.3 V charged capacitor discharges through a 10,000 ohm resistor.

Nominal capacitance $C = 470$ microfarads; resistance $R = 10,000$ ohms.

Expected time constant $\tau = R * C = 4.7$ seconds from these component values.

Procedure: charge, disconnect the supply, close the resistor discharge loop.

Record voltage every 0.25 s from 0 to 20 s; repeat for three trials.

The worksheet includes time_s, voltage_v, trial and temperature_c.

Use all immutable rows for the fit; retain inputs, code and output provenance.

The measurements are a simulation with small, fixed-seed measurement noise.